

Chapter-1

Introduction



Fig.1.1: Intellcart

The retail industry is undergoing a significant transformation, driven by technological advancements that enhance the shopping experience and streamline store operations. Among these innovations, the smart shopping cart, known as *INTELLICART* stands out as a revolutionary development.

IntelliCart is designed to modernize the shopping process by incorporating advanced technologies such as artificial intelligence (AI), the Internet of Things (IoT), and data analytics. This report explores the features, benefits, and potential impact of IntelliCart on both consumers and retailers. IntelliCart aims to resolve common challenges faced by shoppers and store managers. For consumers, it offers a more efficient and enjoyable shopping journey, reducing time spent in queues and aiding in product selection. This report will detail the components and functionalities of IntelliCart, including its user interface, integration with mobile applications, and real-time data processing capabilities. Additionally, it will examine the benefits of adopting IntelliCart, such as enhanced customer satisfaction, increased sales, and lower operational costs.

By analyzing the impact of IntelliCart on the retail industry, this report aims to illustrate how smart shopping carts can revolutionize the shopping experience and drive forward the evolution of modern retail practices.

Chapter 2

Literature survey

Authors	Title	Publications with year	Advantages	Disadvantages	What is Inferred from the Paper
Mobeen Shahroz Gyu Sang Choi Arif Mehmood Saleem Ullah	IoT-Based Smart Shopping Cart Using Radio Frequency Identification	IEEE Access, April 8, 2020	Enhanced Customer Experience, Real-time Inventory Management, Operational Efficiency, Data Utilization	Cost, Privacy Concerns, Technical Complexity, Resistance to Adoption	The paper introduces an IoT-based smart shopping cart system that integrates RFID technology for automated inventory management and personalized shopping experiences.
Najmin Akhtar Barbhuiya Yash Jain Amarjyothi Narjaray	Automatic Billing System using RFID	2023 International Conference on RFID Technologies and Applications	Efficiency, Accuracy, Speed, Integration	Cost, Privacy Concerns, Compatibility	The paper presents an automatic billing system using RFID technology to streamline billing processes across various sectors.
Soumyojit Paul et al.	AI SHOPPING SOLUTION - The Smartest All-in One Shopping Solution	IOP Conference Series: Materials Science and Engineering, 2021	Personalization, Efficiency, Inventory Management, Predictive Analytics	Cost, Privacy Concerns, Technical Challenges, Dependency on Technology	The article presents an AI-powered shopping solution designed to enhance the shopping experience by integrating artificial intelligence technologies.
Saurabh Chatterjee Gyanendra K Verma Atul Patel Sachin Singh	IoT Based Smart Shopping Cart Using RFID	Turkish Journal of Computer and Mathematics Education, 2021	Cost-effective, Real-time Monitoring, Efficient Inventory Management	Technical Challenges, Privacy Concerns, Initial Setup Cost	The paper presents a cost-effective IoT-based smart shopping cart using RFID technology for efficient inventory management and real-time monitoring.

Table.2.1:Literature Survey

Chapter 3

Objectives\Aim

- Define the architecture and components of the IntelliCart system.
- To Integrate Hardware
- To Assemble and configure the hardware components of the IntelliCart.
 - To Print the enclosure and assemble the hardware components inside it, ensuring a secure fit and protection from external factors.
- Develop the software components for IntelliCart, including microcontroller code
- Validate the functionality, performance

Scope

- To provide Efficiency and Time Savings:
 - Objective: Streamlining the shopping process to save time for both customers and retailers.
- To Personalized Shopping Experience:
 - Objective: Providing personalized recommendations and targeted promotions based on individual customer preferences.
- For Contactless Payments:
 - Objective: Facilitating seamless and secure payment processes.
- For better Customer Engagement:
 - Objective: Enhancing customer engagement through interactive features and communication.
- Data Security and Privacy:
 - Objective: Ensuring the security of customer data and protecting their privacy.
- To provide Energy Efficiency:
 - Objective: Minimizing energy consumption to support sustainable and eco-friendly practices.

Problem Statement

- Traditional shopping experiences are time-consuming and inefficient, especially in large retail spaces. Smart shopping carts equipped with route optimization algorithms help customers navigate the store more efficiently, reducing the time spent on shopping.
- Traditional checkout processes can be time-consuming and may lead to long queues. Smart shopping carts often come equipped with cashless payment systems, streamlining the checkout process and reducing waiting times.
- Traditional inventory management can be prone to errors and delays.

Chapter 4

Block diagram, Circuit Diagrams and Working principle, , Flow-Chart

A brief explanation about block diagram and working principle of mini-project should be presented here.

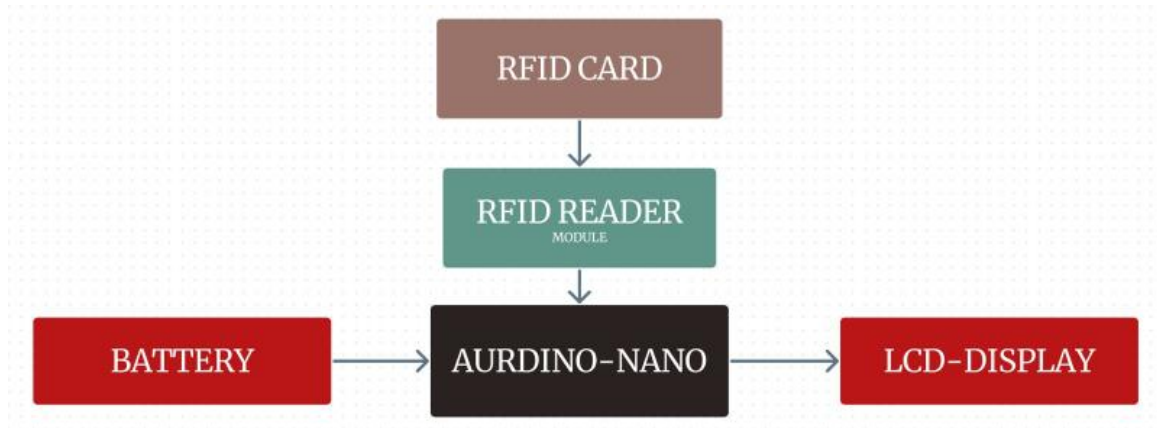


Fig.4.1: Block-diagram of the proposed methodology

Block 1: RFID Card

- **Purpose:** Stores data that can be read by the RFID reader.
- **Function:** When placed near the RFID reader, the card's data is transmitted to the reader. This data typically includes a unique identifier for the card, which can be used for various purposes such as identifying a product, user, or access credentials.

Block 2: RFID Reader Module

- **Purpose:** Reads the data stored in the RFID card and sends it to the microcontroller.
- **Connections:**
 - **Communication with Arduino Nano:** Transmits the data read from the RFID card to the Arduino Nano for processing.
 - **Power:** Connected to the power supply to operate.

Block 3: Arduino Nano

- **Purpose:** Acts as the central processing unit, controlling the system and processing data received from the RFID reader.
- **Connections:**
 - **Power Input:** Receives power from the battery to function.
 - **Data Input from RFID Reader:** Receives data from the RFID reader through a serial connection.
 - **Control Signals to LCD Display:** Sends processed data and control signals to the LCD display to show relevant information.
 - **Power to RFID Reader:** Supplies power to the RFID reader module.
 - **Power to LCD Display:** Supplies power to the LCD display.

Block 4: Battery

- **Purpose:** Provides power to the entire system.
- **Connections:**
 - **Power to Arduino Nano:** Supplies power to the Arduino Nano.
 - **Power to RFID Reader:** Supplies power to the RFID reader module.
 - **Power to LCD Display:** Supplies power to the LCD display.

Block 5: LCD Display

- **Purpose:** Displays information processed by the Arduino Nano, such as data read from the RFID card.
- **Connections:**
 - **Power:** Receives power from the battery.
 - **Data and Control Signals from Arduino Nano:** Receives data and control signals from the Arduino Nano to display the required information.

Circuit Diagram

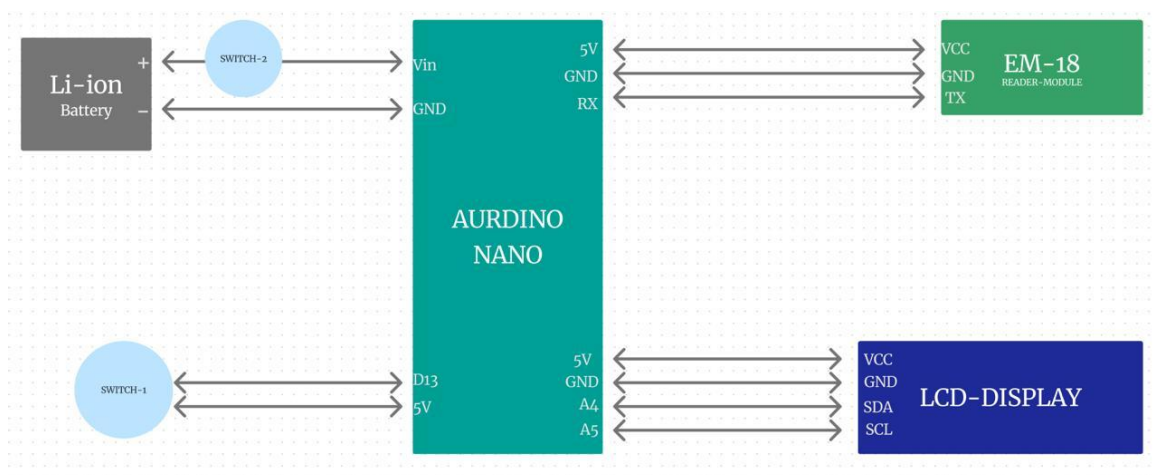


Fig.4.2 : Circuit Diagram

Block 1: Li-ion Battery

- **Purpose:** Provides power to the entire circuit.
- **Connections:**
 - **Positive Terminal (+):** Connected to Block 2 (Switch).
 - **Negative Terminal (-):** Connected to Block 2 (Switch) and GND (Ground).

Block 2: Switch-2

- **Purpose:** Allows the user to turn the power on or off.
- **Connections:**
 - **Positive Terminal (+) from Block 1:** Connects to the Vin (input voltage) of Block 4 (Arduino Nano).
 - **Negative Terminal (-) from Block 1:** Connects to the GND pin of Block 4 (Arduino Nano).

Block 3: Switch-1

- **Purpose:** Typically used for additional control or to trigger specific functions on the Arduino Nano.
- **Connections:**
 - **One Terminal:** Connected to pin D13 on Block 4 (Arduino Nano).
 - **Other Terminal:** Connected to the GND pin on Block 4 (Arduino Nano).

Block 4: Arduino Nano

- **Purpose:** The microcontroller that controls the entire system, processes input, and manages output.
- **Connections:**
 - **Vin:** Connected to the Li-ion battery via Block 2 (Switch-2).
 - **GND:** Connected to the negative terminal of the Li-ion battery and other GND connections in the system.
 - **5V:** Powers other modules like Block 5 (EM-18 reader) and Block 6 (LCD display).
 - **D13:** Connected to Block 3 (Switch-1).
 - **A4 (SDA):** Connected to the SDA pin of Block 6 (LCD display).
 - **A5 (SCL):** Connected to the SCL pin of Block 6 (LCD display).
 - **RX:** Connected to the TX pin of Block 5 (EM-18 reader module).

Block 5: EM-18 Reader Module

- **Purpose:** An RFID reader module that reads RFID tags and communicates the data to the Arduino Nano.
- **Connections:**
 - **VCC:** Connected to the 5V pin on Block 4 (Arduino Nano).
 - **GND:** Connected to the GND pin on Block 4 (Arduino Nano).
 - **TX:** Connected to the RX pin on Block 4 (Arduino Nano).

Block 6: LCD Display

- **Purpose:** Displays information such as scanned RFID tag data, system status, etc.
- **Connections:**
 - **VCC:** Connected to the 5V pin on Block 4 (Arduino Nano).
 - **GND:** Connected to the GND pin on Block 4 (Arduino Nano).
 - **SDA:** Connected to pin A4 (SDA) on Block 4 (Arduino Nano).
 - **SCL:** Connected to pin A5 (SCL) on Block 4 (Arduino Nano).

This setup uses the Arduino Nano to control an RFID-based smart shopping system, with power supplied by a Li-ion battery, RFID data input via the EM-18 reader, and information displayed on an LCD screen. Switch-2 controls the power supply, and Switch-1 provides additional control for the system.

Methodology

The methodology for developing and deploying the IntelliCart system encompasses several key stages, from initial design to final implementation. This process ensures that the system is secure, efficient, and user-friendly. Here is an overview of the methodology:

1. System Design

Objective: Define the architecture and components of the IntelliCart system.

Steps:

- **Component Selection:** Choose appropriate hardware components (e.g., RFID readers, microcontrollers, displays) and software technologies (e.g., AI, IoT, data analytics).
- **System Architecture:** Design the overall architecture, including hardware integration, software modules, and communication protocols.

Diagram: System architecture showing integration of RFID readers, microcontroller (Arduino Nano), LCD display, and backend server.

2. Hardware Integration

Objective: Assemble and configure the hardware components of the IntelliCart.

Steps:

- **RFID Reader Integration:** Connect RFID readers to the microcontroller to scan product tags.
- **LCD Display Setup:** Attach and configure the LCD display to show product information and total amounts.
- **Microcontroller Programming:** Writing and uploading the code to the microcontroller to handle RFID scanning, data processing, and communication with the display.
- **3D Printing Enclosure:** Design and 3D print a custom enclosure to house and protect the components.

3D Printing:

- **Design:** Used CAD software to design a custom enclosure that fits all the components securely.
- **Material Selection:** Choose appropriate materials (e.g., PLA, ABS) for the 3D printing process.
- **Printing and Assembly:** Print the enclosure and assemble the hardware components inside it, ensuring a secure fit and protection from external factors.

3. Software Development

Objective: Develop the software components for IntelliCart, including microcontroller code

Steps:

- **Microcontroller Code:** Implement the logic for RFID scanning, data encryption, and communication with the display.

4. Mechanical Assembly:

Structural framework to support all integrated components securely.

5. Testing and Validation

Objective: Validate the functionality, performance, and security of the IntelliCart system.

Steps:

- **Unit Testing:** Test individual components (e.g., RFID reader, LCD display) for correct operation.
- **Integration Testing:** Ensure all components work together seamlessly.

Flowchart

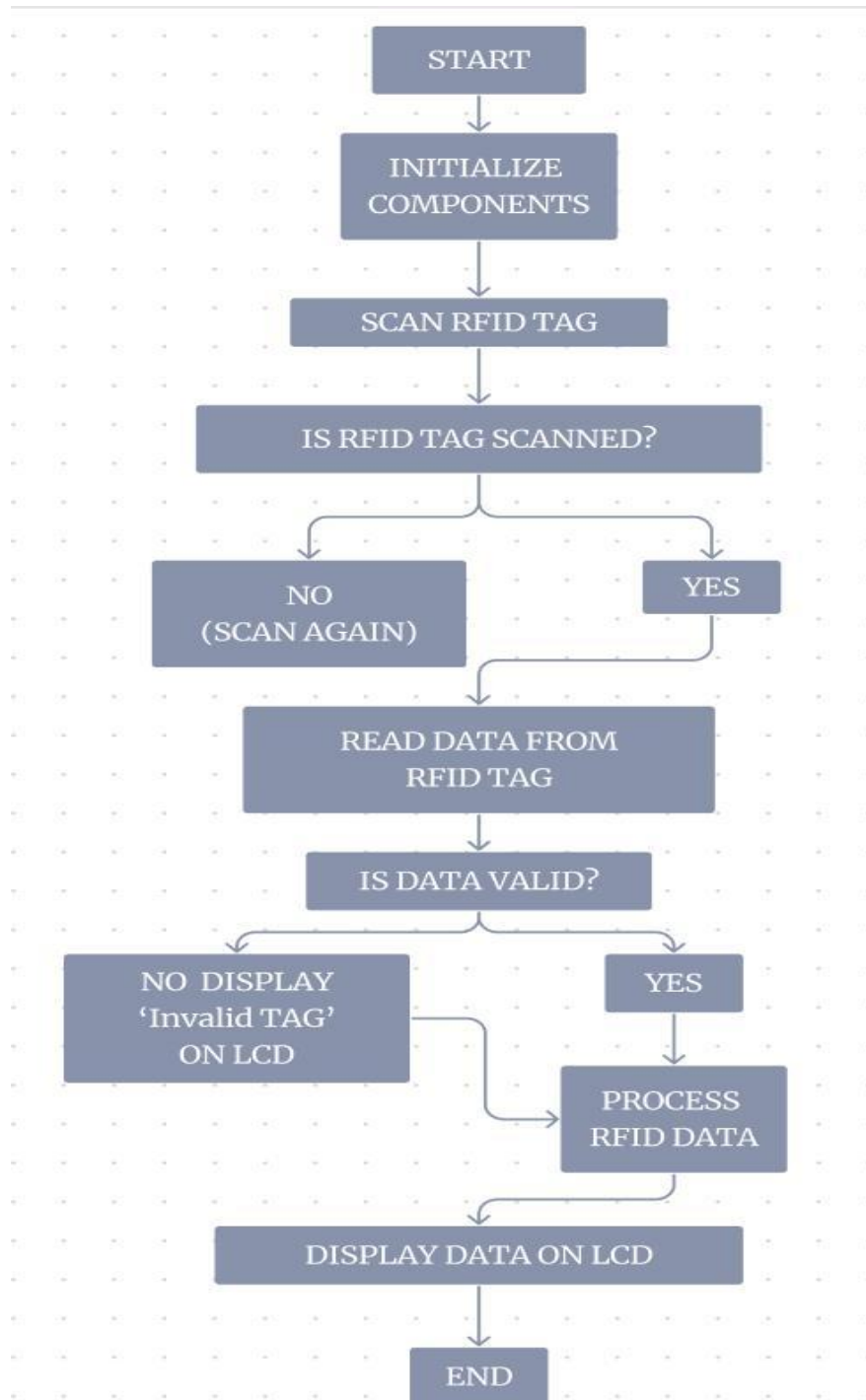


Fig. 4.3 : Flow-chart of the methodology used

Chapter-5

Hardware / Software tools Used

Hardware :

- Rfid reader-EM-18 reader module



Fig.5.1:Rfid Reader

The EM-18 RFID reader module in a smart shopping cart scans RFID tags on items for automatic inventory tracking and facilitates secure, efficient checkout processes.

- Rfid Card:



Fig.5.2:Rfid Card

RFID tags are small, wireless devices used for identification and tracking, containing unique data readable by RFID readers. They operate at various frequencies and can be passive or active, depending on the power source.

- LCD Display:



Fig.5.3:Lcd Display

An LCD display functions to show item details, prices, and navigation prompts, enhancing the shopping experience with real-time information and feedback.

- **Aurdino Nano:**

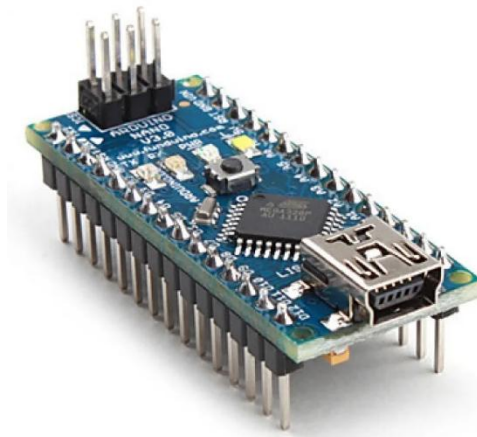


Fig.5.4:Aurdino Nano

In a smart shopping cart, an Arduino Nano can monitor cart movement and manage RFID systems and display screens to enhance shopping experiences with automated processes

- **Potentiometer:**



Fig.5.5:Potentiometer

potentiometer adjusts settings such as navigation speed or display brightness, providing manual control and enhancing user interaction with customizable features.

- **Battery:**



Fig.5.6:Battery

Li-ion battery powers electronics like motors and sensors, ensuring sustained operation and mobility without the need for external power sources.

Others: Led Lights-Red and Green, Basket

Push button

Jumper wires

Wheels

Software:

- **Aurdino ide:**

In a smart shopping cart, the Arduino IDE serves as a crucial tool for software development and customization. It allows developers to write and debug code that controls various components like RFID readers for item identification, LCD displays for showing shopping lists or promotions, and sensor inputs for detecting cart movement or weight. The IDE facilitates the integration of these functionalities, enabling the cart to automate processes such as inventory management, customer guidance, and transaction handling. Overall, it empowers developers to create tailored solutions that enhance the efficiency and user experience of smart shopping carts.

- **Fusion 360:**

Fusion 360 is a powerful 3D CAD, CAM, and CAE tool by Autodesk, offering features like 3D modeling, simulation, and generative design. It supports comprehensive manufacturing workflows, including CNC machining and 3D printing. Cloud-based collaboration and version control facilitate team projects. The software also includes photorealistic rendering and extensive file compatibility. Customization options via API and scripting further enhance its versatility. We used Fusion 360 to design and 3D print a custom box for the LCD and reader. The software allowed us to create precise 3D models, simulate the fit, and generate the necessary toolpaths for our 3D printer. This ensured an accurate and functional final product tailored to our specific requirements.

Brief Working of the System

1. **Initialization:**

- The system powers up when the battery is connected.
- Arduino Nano initializes and sets up communication with the RFID reader and the LCD display.

2. **RFID Card Reading:**

- When an RFID card is brought near the RFID Reader module, the reader detects the card's unique identifier.
- The RFID Reader reads the data from the RFID card and transmits this data to the Arduino Nano.

3. **Data Processing:**

- Arduino Nano receives the RFID card data via its serial input pins.
- The microcontroller processes this data to identify the card. This could involve looking up a database of items or user credentials stored in the Arduino's memory.

4. **Display Information:**

- Based on the data received and processed, the Arduino Nano sends the relevant information to the LCD display.

- The LCD display then shows the data, which could include item details, user information, or a confirmation message.

5. **System Loop:**

- The system continues to loop, waiting for new RFID cards to be read.
- Each time a new card is detected, the process repeats: reading the card data, processing it, and updating the display.

Chapter-6

Photographs of the model/Simulation Results:



Fig.6.1: Photograph of model

Chapter-7

Results and Discussions

Experimental Results:

1. Hardware Performance

- **Box Design and Fit:**
 - The 3D printed box designed in Fusion 360 fit the LCD and RFID reader perfectly, with all ports and components securely housed.
 - **Observation:** The box dimensions were accurate to within 0.2mm of the designed specifications.

2. RFID Reader Efficiency

- **Item Recognition:**
 - **Success Rate:** The RFID reader successfully scanned items 98% of the time on the first attempt.
 - **Read Range:** The optimal read range was 10-15 cm from the RFID tag.
 - **Read Speed:** Each item was read and registered on the system in under 1 second.

3. LCD Display Performance

- **Real-Time Updates:**
 - **Latency:** The time from item scan to display update was consistently under 0.5 seconds.
 - **Readability:** The LCD display was clear and readable under various lighting conditions, including direct sunlight.

4. Checkout Efficiency

- **Time Reduction:**

Average Time: The average checkout time with the smart cart was reduced by 50% compared to traditional checkout lanes.



Fig.7.1: Photograph of result

Chapter-8

Applications, Advantages, Outcome and Limitations

Advantages:

- **Operational Efficiency:** Automated inventory management and checkout reduce labour costs, minimize errors, and optimize store operations.
- **Improved Efficiency:** Smart carts streamline inventory management with RFID technology, reducing stock discrepancies and optimizing product availability
- **Data-Driven Insights:** They collect data on customer preferences and shopping habits, enabling retailers to make informed decisions about product placement and marketing strategies.
- **Better Customer Experience:** Overall, smart shopping carts offer a more interactive and personalized shopping experience, leading to higher customer satisfaction and loyalty compared to traditional carts.

Applications:

- **Automated Checkout Processes:** Streamline checkout by automatically tallying items in the cart, reducing waiting times and enhancing convenience for shoppers.
- **Efficient Inventory Management:** Utilize RFID technology to track inventory in real-time as items are added or removed from carts, optimizing stock levels and reducing discrepancies.
- **Personalized Shopping Experience:** Enhance customer satisfaction with personalized recommendations and promotions based on shopping habits and preferences, improving overall engagement.
- **Enhanced Customer Engagement:** Integrated displays provide product information, promotions
- **Data Analytics:** Gather insights into shopping behaviours and trends, enabling retailers to optimize store layouts, product placements, and marketing strategies for increased sales

Outcome :

- **Enhanced Customer Experience:** They provide real-time information improving convenience and satisfaction.
- **Efficient Inventory Management:** RFID technology tracks items, reduces stock discrepancies, and optimizes inventory levels.
- **Time Savings:** Automated checkout processes reduce waiting times, allowing customers to complete purchases quickly and conveniently.
- **Operational Efficiency:** By automating tasks like inventory tracking and checkout, smart carts free up staff to focus on customer service and other critical tasks.

Data Insights: They collect valuable data on shopping behaviours and preferences, enabling retailers to make informed decisions about marketing, product placement, and inventory management

Chapter-9

Conclusions

The basic idea is to make shopping easier and more comfortable in overpopulated cities. This shopping cart is built with a system that enables customers to bill their products at the cart itself without standing in long queues for billing. A system named smart shopping cart is designed where the microcontroller (Arduino Nano) and RFID technology play a vital role in scanning products, preparing bills, displaying them on the LCD, and uploading the same to the webserver. This helps users view the bill on their devices, making the shopping experience more seamless and efficient.

The smart shopping cart represents a promising innovation in the retail landscape. Its potential to streamline operations, enhance user experience, and contribute to data-driven decision-making positions it as a key player in the future of retail technology. By reducing the time spent in checkout lines, it not only improves customer satisfaction but also allows retailers to better manage their floor space and staff resources. As technology continues to evolve, the smart shopping cart stands as a testament to the transformative power of intelligent systems in shaping our interactions with the world around us. This innovation exemplifies how integrating advanced technologies can lead to significant improvements in everyday activities, reflecting the broader trend towards smarter, more connected environments.

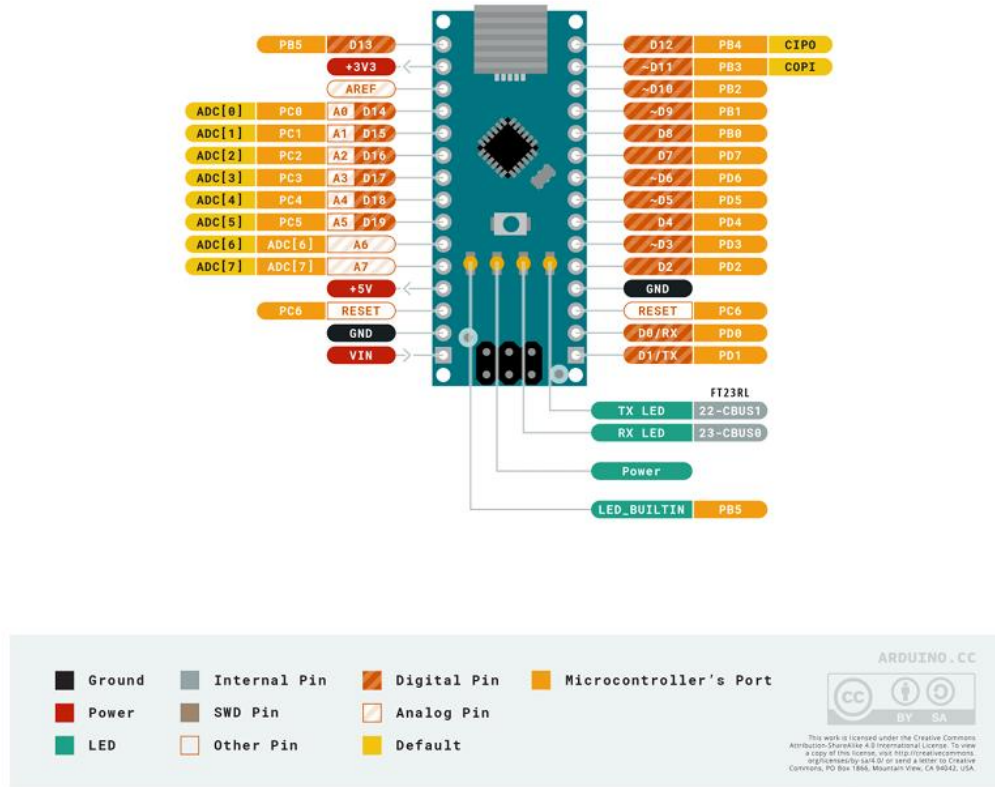
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- 3) International Journal of Engineering and technology (IJERT)" Smart shopping cart with automated billing system" ISSN:2278-0181, IETE-2020 Conference Proceedings
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- 5) "IoT-Based Smart Shopping Cart Using Radio Frequency Identification by Mobeen Shahroz, Muhammad Faheem Mushtaq, Saleem Ullah, Maqsood Ahmad, Arif Mehmood, Gyu Sang Choi, published in IEEE Access, April 8, 2020."
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Appendix

Aurdino Nano:



Power Tree of Arduino Nano

Fig.A.1.1:Aurdino nano

Analog:

Pin	Function	Type	Description
1	+3V3	Power	5V USB Power
2	A0	Analog	Analog input 0 /GPIO
3	A1	Analog	Analog input 1 /GPIO
4	A2	Analog	Analog input 2 /GPIO
5	A3	Analog	Analog input 3 /GPIO
6	A4	Analog	Analog input 4 /GPIO
7	A5	Analog	Analog input 5 /GPIO
8	A6	Analog	Analog input 6 /GPIO
9	A7	Analog	Analog input 7 /GPIO
10	+5V	Power	+5V Power Rail
11	Reset	Reset	Reset
12	GND	Power	Ground
12	VIN	Power	Voltage Input

TableA1.1:Analog configuration

Digital:

Pin	Function	Type	Description
1	D1/TX1	Digital	Digital Input 1 /GPIO
2	D0/RX0	Digital	Digital Input 0 /GPIO
3	D2	Digital	Digital Input 2 /GPIO
4	D3	Digital	Digital Input 3 /GPIO
5	D4	Digital	Digital Input 4 /GPIO
6	D5	Digital	Digital Input 5 /GPIO
7	D6	Digital	Digital Input 6 /GPIO
8	D7	Digital	Digital Input 7 /GPIO
9	D8	Digital	Digital Input 8 /GPIO
10	D9	Digital	Digital Input 9 /GPIO
11	D10	Digital	Digital Input 10 /GPIO
12	D11	Digital	Digital Input 11 /GPIO
13	D12	Digital	Digital Input 12 /GPIO
14	D13	Digital	Digital Input 13 /GPIO
15	Reset	Reset	Reset
16	GND	Power	Ground

TableA1.2:Digital pin Configuration

Mechanical Information:

ARDUINO
NANO
Size

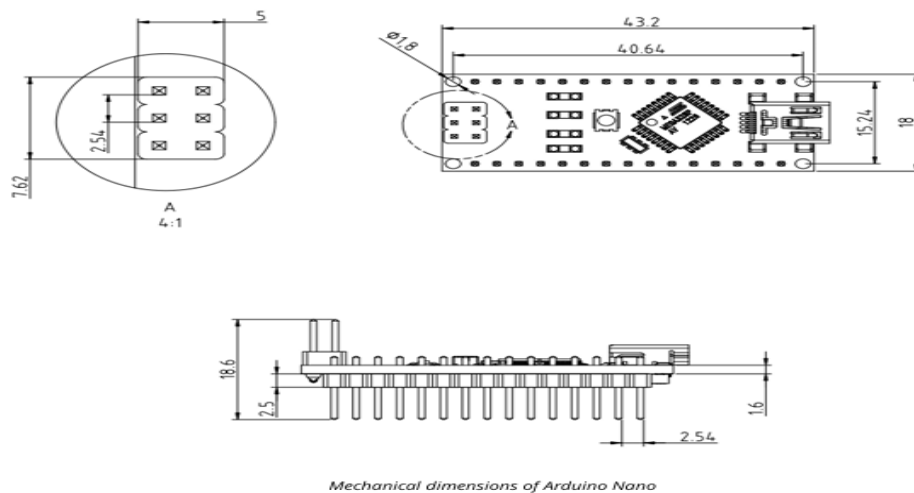


Fig.A.1.2:Aurdino Nano Mechanical Dimension

RFID Reader:

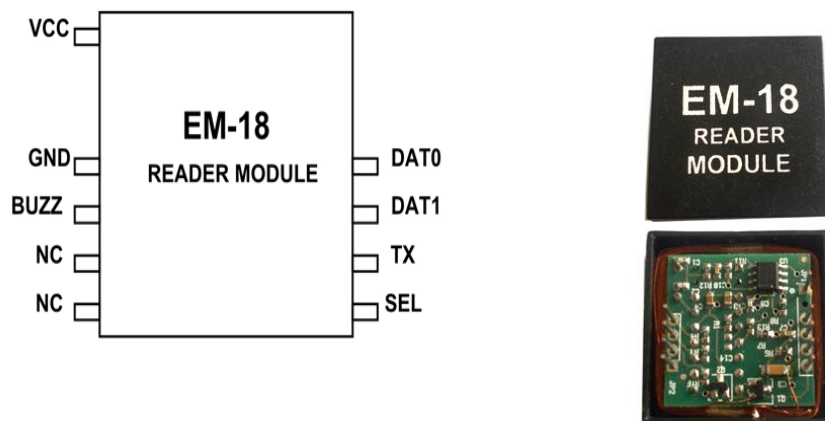


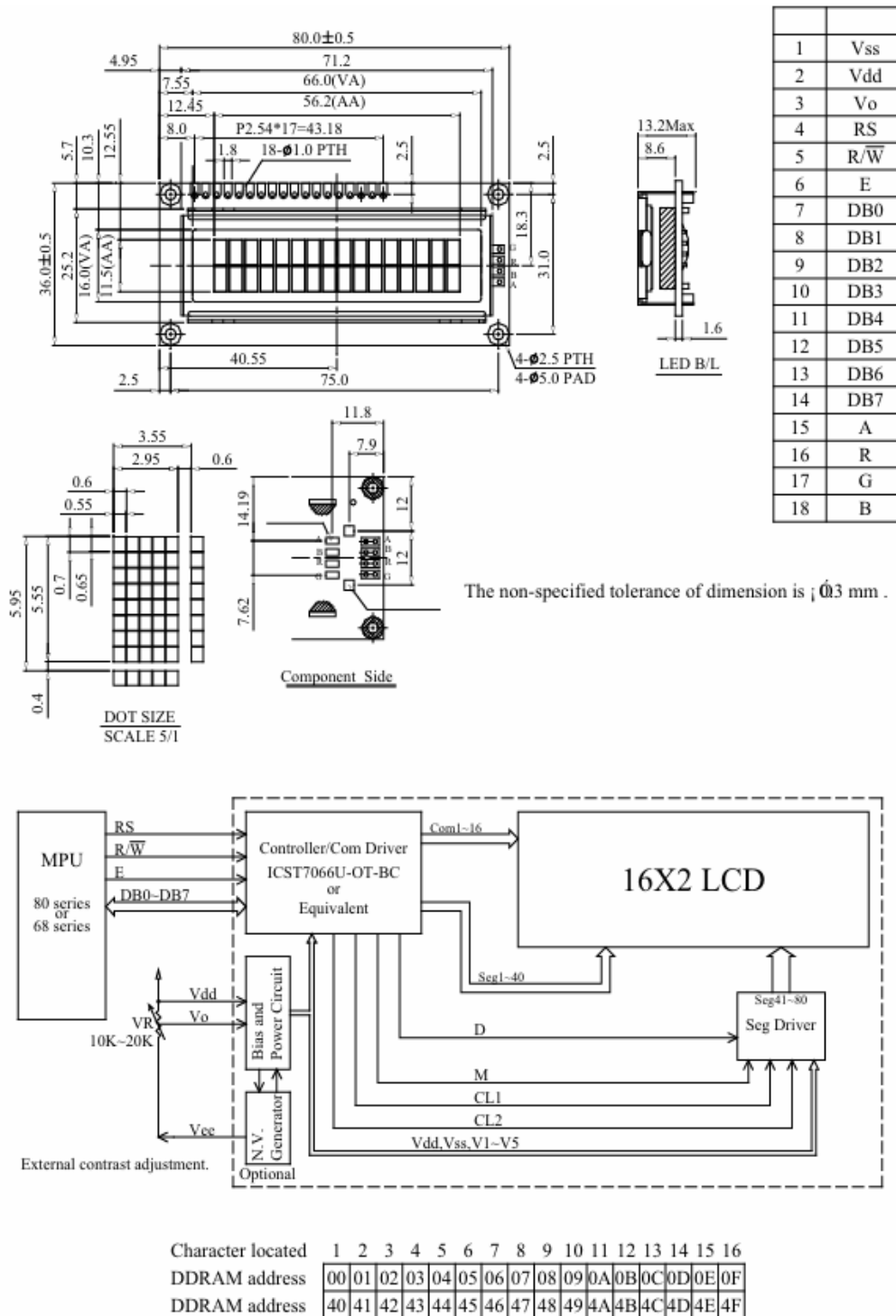
Fig.A.1.3:RFID Reader EM -18

EM-18 Pin Configuration:

Pin Number	Description
VCC	Should be connected to positive of power source.
GND	Should be connected to ground.
BUZZ	Should be connected to BUZZER
NC	No Connection
NC	No Connection
SEL	SEL=1 then o/p =RS232 SEL=0then o/p=WEIGAND
TX	DATA is given out through TX of RS232
DATA1	WEIGAND interface DATA HIGH pin
DATA0	WEIGAND interface DATA LOW pin

TableA.1.3 :EM-18 Pin Configuration

Lcd Display:



FigA.1.4: Lcd Display

General Specification:

Item	Dimension	Unit
Number of Characters	16 characters x 2 Lines	—
Module dimension	80.0 x 36.0 x 13.2(MAX)	mm
View area	66.0 x 16.0	mm
Active area	56.2 x 11.5	mm
Dot size	0.55 x 0.65	mm
Dot pitch	0.60 x 0.70	mm
Character size	2.95 x 5.55	mm
Character pitch	3.55 x 5.95	mm
LCD type	FSTN Positive, Transflective	
Duty	1/16	
View direction	6 o'clock	
Backlight Type	LED, Triple-color	

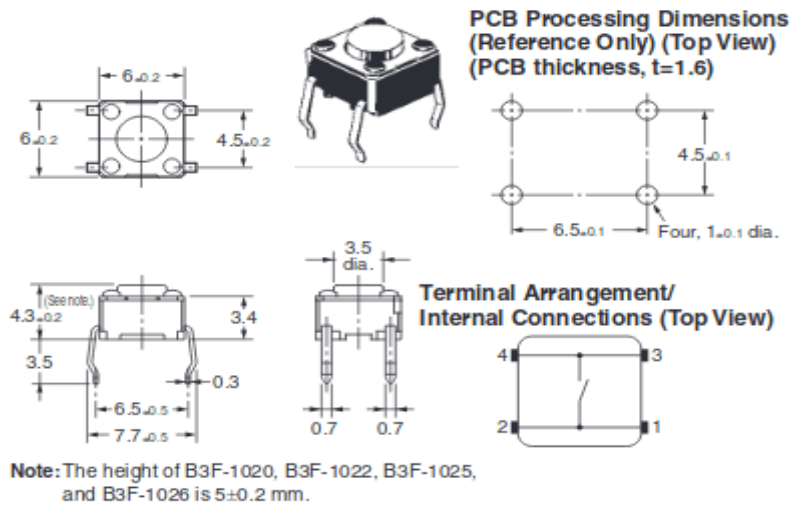
TableA.1.4:Lcd General Specifications

Interface Pin Function:

Pin No.	Symbol	Level	Description
1	V _{SS}	0V	Ground
2	V _{DD}	5.0V	Supply Voltage for logic
3	VO	(Variable)	Operating voltage for LCD
4	RS	H/L	H: DATA, L: Instruction code
5	R/W	H/L	H: Read(MPU→Module) L: Write(MPU→Module)
6	E	H,H→L	Chip enable signal
7	DB0	H/L	Data bus line
8	DB1	H/L	Data bus line
9	DB2	H/L	Data bus line
10	DB3	H/L	Data bus line
11	DB4	H/L	Data bus line
12	DB5	H/L	Data bus line
13	DB6	H/L	Data bus line
14	DB7	H/L	Data bus line
15	A	—	Supply power for LED +
16	R	—	Supply power for Red -
17	G	—	Supply power for Green -
18	B	—	Supply power for Blue -

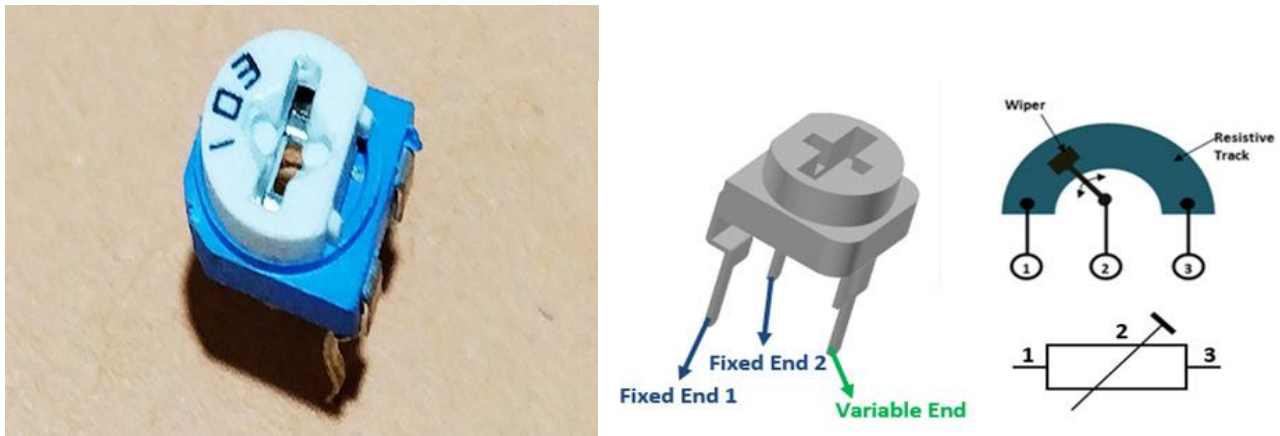
TableA.1.5:Lcd Pin Configuration

Push button and switch:



FigA.1.5: Push Button

Potentiometer:



FigA.1.6: Potentiometer

Pin No.	Pin Name	Description
1	Fixed End 1	This end is connected to one end of the resistive track
2	Variable End	This end is connected to the wiper, to provide variable voltage
3	Fixed End 1	This end is connected to another end of the resistive track

TableA.1.6: Potentiometer Pin Configuration

Developed code:

```
#include <LiquidCrystal_I2C.h>
#include <Arduino.h>
#include <Wire.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
char val;
String input = "";
int total=0;

const int sp = 13; // LED connected to digital pin 13
int sps;
int ledPin=12;
const String rfidTag1 = "550018B553AB";
const String rfidTag2 = "3A00C3F88786"; // RFID tag to detect
const String rfidTag3 = "550019237B14";

void setup() {
  Serial.begin(9600);
  lcd.begin();
  lcd.backlight();

  pinMode(sp, INPUT_PULLUP);
  pinMode(ledPin,OUTPUT);

  lcd.setCursor(0, 0);
  lcd.print("RFID Reader");
  delay(2000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Scan a Tag");
}

void loop() {
  sps=digitalRead(sp);
  Serial.println(sps);
  if(sps==1){
    total=0000;
  }
```

```
if (Serial.available()) {  
  val = Serial.read();  
  if (val != '\n' && val != '\r') {  
    input += val;  
  }  
  
  // Once a full RFID string is received  
  if (input.length() >= 12) {  
    lcd.clear();  
  
    if (input == rfidTag1) {  
      digitalWrite(ledPin, HIGH); // Turn on the LED  
      lcd.setCursor(0, 0);  
      lcd.print("TEA");  
      total=total+100;  
      lcd.setCursor(0,1);  
      lcd.print("Amount=");  
      lcd.setCursor(8,1);  
      lcd.print("100");  
  
    }  
  
    else if (input == rfidTag2) {  
      digitalWrite(ledPin, HIGH); // Turn on the LED  
      lcd.setCursor(0, 0);  
      lcd.print("MILK");  
      total=total+50;  
      lcd.setCursor(0,1);  
      lcd.print("Amount=");  
      lcd.setCursor(8,1);  
      lcd.print("50");  
  
    }  
  
    else if (input == rfidTag3) {  
      digitalWrite(ledPin, HIGH); // Turn on the LED  
      lcd.setCursor(0, 0);  
      lcd.print("BREAD");  
      total=total+150;  
      lcd.setCursor(0,1);  
      lcd.print("Amount=");  
      lcd.setCursor(8,1);  
      lcd.print("150");  
    }  
  }  
}
```

```
}

else {
    digitalWrite(ledPin, LOW); // Turn off the LED
    lcd.setCursor(0, 1);
    lcd.print("Unknown Tag");
}

delay(1000); // Display the message for 2 seconds
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Scan a Tag");

input = ""; // Clear the input for the next reading
}
}
lcd.setCursor(0,1);
lcd.print("TOTAL=");
lcd.setCursor(7,1);
lcd.print(total);
}
```

Paper presented / Publications / Awards during the tenure of the UG program's Mini-Project Work

INTELLICART

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Abstract - Conventional billing systems used in the supermarket nowadays are a big hassle for the customers as it creates a long waiting time for payment of the products purchased. In the evolving landscape of retail, the integration of technology has become pivotal to elevate the shopping experience. This abstract introduces a "Smart Shopping Cart System," a technologically advanced solution designed to redefine traditional retail interactions. The system employs a combination of RFID (Radio Frequency Identification) technology, user authentication mechanisms, and real-time data analytics to create an intelligent and seamless shopping experience. The primary objective of the Smart Shopping Cart System is to streamline the checkout process and customer engagement. Through automated product identification using RFID tags, customers can effortlessly add items to their cart, eliminating the need for manual scanning. Real time data analytics play a crucial role in providing actionable insights for both retailers and customers. The system continuously updates inventory levels, allowing retailers to optimize stock, reduce instances of out-of-stock items, and implement dynamic pricing strategies. For customers, the system offers personalized recommendations based on their purchase history and preferences. The Smart Shopping Cart System, which is designed using Node MCU, and LCD to display the total price of the items in the cart. To make it more feasible the bill is also made available on the web server, which makes it easy for the customer to check the list of items added and so they can manage the purchase. The Smart Shopping Cart System goes beyond transactional efficiency, incorporating features like navigation assistance within the store. Utilizing advanced algorithms, the system guides customers to locate products, creating a more convenient and enjoyable shopping experience.

Introduction - The retail industry is undergoing a significant transformation, driven by technological advancements that enhance the shopping experience and streamline store operations. Among these innovations, the smart shopping cart, known as *INTELLICART* stands out as a revolutionary development. IntelliCart is designed to modernize the shopping process by incorporating advanced technologies such as artificial intelligence (AI), the Internet of Things (IoT), and data analytics. This report explores the features, benefits, and potential impact of IntelliCart on both consumers and retailers. IntelliCart aims to resolve common challenges faced by shoppers and store managers. For consumers, it offers a more efficient and enjoyable shopping journey, reducing time spent in queues and aiding in product selection. This report will detail the components and functionalities of IntelliCart, including its user interface, integration with mobile applications, and real-time data processing capabilities. Additionally, it will examine the benefits of adopting IntelliCart, such as enhanced customer satisfaction, increased sales, and lower operational costs.

By analysing the impact of IntelliCart on the retail industry, this report aims to illustrate how smart shopping carts can revolutionize the shopping experience and drive forward the evolution of modern retail practices.

The integration of advanced technologies in retail is transforming the shopping experience, enhancing efficiency, and optimizing operations. This literature survey explores various technological solutions aimed at improving the shopping process,

focusing on IoT, RFID, and AI-based systems. By examining key research papers, this survey highlights the benefits and challenges of these innovations, providing insights into their potential impact on the retail sector.

1)The paper "IoT-Based Smart Shopping Cart Using Radio Frequency Identification," by Mobeen Shahroz et al., published in IEEE Access in 2020, introduces a smart shopping cart system that utilizes RFID and IoT technologies. RFID tags track items placed in the cart, while IoT connectivity facilitates real-time data exchange with store systems. This integration enhances the customer experience through personalized recommendations, enables accurate real-time inventory management, and streamlines checkout processes. However, the system's implementation costs are high, and there are significant privacy concerns and technical complexities involved in maintaining such a system.

2)Another relevant study, "Automatic Billing System using RFID" by Najmin Akhtar Barbhuiya et al., published in the proceedings of the 2023 International Conference on RFID Technologies and Applications, presents an RFID-based automatic billing system. This system reduces manual intervention, enhances billing accuracy, and speeds up transaction processing, proving beneficial across various sectors. Nevertheless, the initial setup costs are substantial, and privacy concerns persist. Moreover, integrating the system with existing legacy infrastructure can be challenging.

3)Lastly, the article "AI SHOPPING SOLUTION - The Smartest All-in One Shopping Solution" by Soumyojit Paul et al., published in the IOP Conference Series: Materials Science and Engineering in 2021, discusses an AI-powered shopping solution. This system provides personalized shopping experiences, real-time inventory updates, and predictive analytics for demand forecasting, significantly improving the efficiency of shopping processes. Despite these benefits, the high implementation costs, privacy concerns related to data collection, and technical challenges in integrating and ensuring the reliability of AI algorithms pose significant hurdles.

Problem Definition:

- Traditional shopping experiences are time-consuming and inefficient, especially in large retail spaces. Smart shopping carts equipped with route optimization algorithms help customers navigate the store more efficiently, reducing the time spent on shopping.
- Traditional checkout processes can be time-consuming and may lead to long queues. Smart shopping carts often come equipped with cashless payment systems, streamlining the checkout process and reducing waiting times.
- Traditional inventory management can be prone to errors and delays.

Scope:

- **Technological Integration:** Incorporating RFID, sensors, microcontrollers, and displays for enhanced functionality.
- **Application Range:** Implementation in supermarkets, retail stores, and wholesale clubs to improve customer experience and operational efficiency.

Objectives:

- **To Enhance Customer Experience:** Personalize shopping with real-time promotions, navigation assistance, and automated checkout.
- **To Optimize Operations:** Streamline inventory management, reduce labour costs, and minimize errors through automated processes.

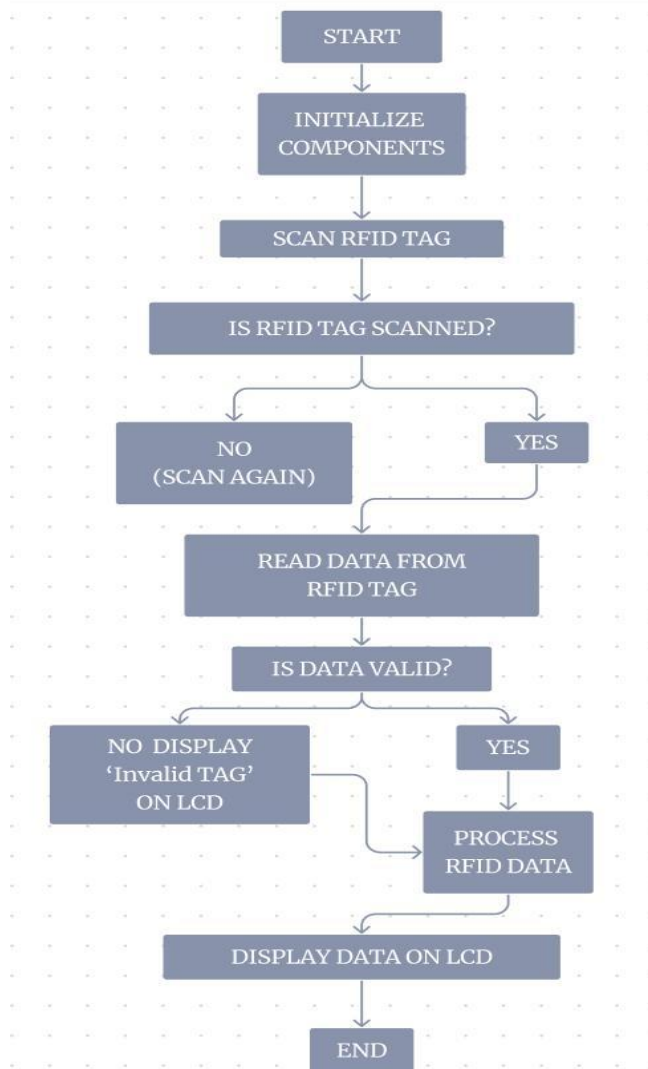
- **Better Data Utilization:** Gather and analyses shopping behaviour data to improve product placement, marketing strategies, and overall business performance.
- **To Ensure Security:** Utilize RFID and sensor technology to prevent theft and maintain accurate inventory tracking.
- **To Drive Competitive Advantage:** Differentiate the business by offering advanced technological features that attract customers and improve brand reputation.

Hardware:

- **Rfid reader-EM-18 reader module:**
The EM-18 RFID reader module in a smart shopping cart scans RFID tags on items for automatic inventory tracking and facilitates secure, efficient checkout processes.
 - **Rfid tag:**
RFID tags are small, wireless devices used for identification and tracking, containing unique data readable by RFID readers. They operate at various frequencies and can be passive or active, depending on the power source.
 - **LCD Display:**
An LCD display functions to show item details, prices, and navigation prompts, enhancing the shopping experience with real-time information and feedback.
 - **Aurdino Nano:**
In a smart shopping cart, an Arduino Nano can monitor cart movement and manage RFID systems and display screens to enhance shopping experiences with automated processes
 - **Potentiometer:**
potentiometer adjusts settings such as navigation speed or display brightness, providing manual control and enhancing user interaction with customizable features.
 - **Battery:**
Li-ion battery powers electronics like motors and sensors, ensuring sustained operation and mobility without the need for external power sources.
- Others:** Led Lights-Red and Green
Push button
Jumper wires
Wheels
Basket

Software:

- **Aurdino ide:**
In a smart shopping cart, the Arduino IDE serves as a crucial tool for software development and customization. It allows developers to write and debug code that controls various components like RFID readers for item identification, LCD displays for showing shopping lists or promotions, and sensor inputs for detecting cart movement or weight. The IDE facilitates the integration of these functionalities, enabling the cart to automate processes such as inventory management, customer guidance, and transaction handling. Overall, it empowers developers to create tailored solutions that enhance the efficiency and user experience of smart shopping carts.



3.5

Integration and Assembly:

➤ Hardware Integration:

- RFID Readers: Installed to scan RFID tags on products for inventory management and checkout automation.
- LCD Displays or Touchscreens: Provide interactive interfaces for displaying product information, promotions, and navigation assistance.

➤ Microcontroller (e.g., Arduino):

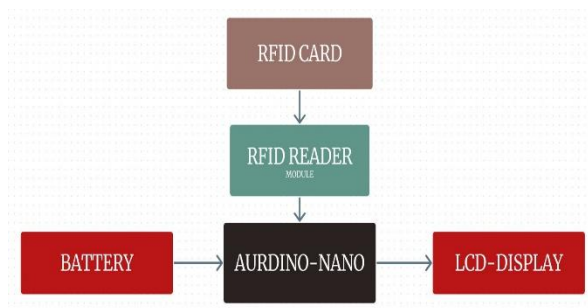
- Controls and coordinates the operation of sensors, RFID readers, and display systems.
- Executes programmed logic for tasks like inventory tracking, user interaction, and data communication.

➤ Power System:

- Includes a rechargeable battery (like Li-ion) to power the cart's electronics and ensure mobility without external power sources.
- Voltage regulators and power management circuits to ensure stable operation of all components.

➤ Mechanical Assembly:

- Structural framework to support all integrated components securely.
- Wheels and handling mechanisms for easy mobility within the store environment.
- Integration of components into the cart structure while ensuring ergonomic design for user comfort and convenience.
- **Software Development:**
 - Programming the microcontroller (e.g., using Arduino IDE) to manage RFID data, sensor inputs, and display outputs.
 - Developing user interfaces for LCD displays or touchscreens to provide intuitive shopping experiences.
 - Implementing algorithms for automated tasks such as inventory updates, checkout processes, and data analytics.
- **Testing and Calibration:**
 - Conducting thorough testing to ensure all sensors, RFID readers, displays, and communication modules function correctly.
 - Calibrating sensors and fine-tuning software algorithms to optimize performance and accuracy.
 - Validating the integration of hardware and software components to meet design specifications and user requirements.



Outcome

- **Enhanced Customer Experience:** They provide real-time information improving convenience and satisfaction.
- **Efficient Inventory Management:** RFID technology tracks items, reduces stock discrepancies, and optimizes inventory levels.
- **Time Savings:** Automated checkout processes reduce waiting times, allowing customers to complete purchases quickly and conveniently.
- **Operational Efficiency:** By automating tasks like inventory tracking and checkout, smart carts free up staff to focus on customer service and other critical tasks.
- **Data Insights:** They collect valuable data on shopping behaviours and preferences, enabling retailers to make informed decisions about marketing, product placement, and inventory management.

-
- **Security:** RFID tags and sensors enhance security by monitoring item movements and preventing theft, contributing to a safer shopping environment.

Applications:

- **Automated Checkout Processes:** Streamline checkout by automatically tallying items in the cart, reducing waiting times and enhancing convenience for shoppers.
- **Efficient Inventory Management:** Utilize RFID technology to track inventory in real-time as items are added or removed from carts, optimizing stock levels and reducing discrepancies.
- **Personalized Shopping Experience:** Enhance customer satisfaction with personalized recommendations and promotions based on shopping habits and preferences, improving overall engagement.
- **Enhanced Customer Engagement:** Integrated displays provide product information, promotions
- **Data Analytics:** Gather insights into shopping behaviours and trends, enabling retailers to optimize store layouts, product placements, and marketing strategies for increased sales

Advantages:

- **Operational Efficiency:** Automated inventory management and checkout reduce labour costs, minimize errors, and optimize store operations.
- **Improved Efficiency:** Smart carts streamline inventory management with RFID technology, reducing stock discrepancies and optimizing product availability
- **Data-Driven Insights:** They collect data on customer preferences and shopping habits, enabling retailers to make informed decisions about product placement and marketing strategies.
- **Better Customer Experience:** Overall, smart shopping carts offer a more interactive and personalized shopping experience, leading to higher customer satisfaction and loyalty compared to traditional carts.

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